

Can cosmology confirm that dark energy is a cosmological constant? An observable-subspace bound

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Abstract

The simplest hypothesis for dark energy is that it is vacuum energy—a cosmological constant with equation of state $w = -1$ and no perturbations. We ask a precise question: given a finite suite of cosmological probes, which deviations from a cosmological constant are observable even in principle? We formalise the observability of a dark-energy feature to a probe as the norm of the feature’s linear response projected onto the probe’s sensitivity subspace, $\|\Pi_{\text{probe}} \mathcal{R}\|$, and show that the three questions answered separately in a companion programme—the detectability of a localised background feature, the discrimination of horizon-entropy models, and the closure of the structure-growth channel—are one construction under three projections. The finiteness of any real probe suite then implies a sharp epistemic bound: the observable part of the dark-energy response space is the span of the probe kernels, and its orthogonal complement is invisible in principle. Using physically-motivated kernels for six probes, we find that the cosmological-constant hypothesis *is* falsifiable in the directions the data cover—low-redshift background evolution and any clustering are well observed—but that a structurally-defined family of near-constant alternatives, deviating only at high redshift, on super-horizon scales, or in a mid-redshift observational desert, is indistinguishable from vacuum energy. The strongest statement current cosmology can support is therefore “dark energy is a cosmological constant within the observable subspace,” never without qualification. We map the blind family explicitly.

1 Introduction

The most economical account of cosmic acceleration is that the dark energy driving it is the energy of the vacuum: a cosmological constant Λ with equation of state $w = -1$ exactly and, being Lorentz invariant, no spatial perturbations at all. Every current observation is consistent with this account. It is also the account with the least explanatory content—it names the substance without addressing why its density takes the value it does—and so a large literature seeks dynamical alternatives whose deviations from $w = -1$ might be detected.

This paper asks a question logically prior to that search: setting aside whether a given experiment has the sensitivity, *which* deviations from a cosmological constant are observable *even in principle* by a finite suite of probes? The question is not rhetorical. A cosmological constant is a single point in an infinite-dimensional space of possible dark-energy behaviours—a density that could in principle depend on scale and time in arbitrarily many ways—and a finite set of experiments can constrain only a finite-dimensional projection of that space. The complement is not merely poorly measured; it is inaccessible to those probes for any exposure.

We make this precise and find that it has teeth. The observability of a dark-energy feature reduces to a single object, and the various detectability and discrimination results established elsewhere in this programme are its instances. The finiteness of the probe suite then yields an explicit observable subspace and an explicit blind complement. Applied to the cosmological-constant hypothesis, the result is neither the complacent “ Λ fits, so dark energy is vacuum energy” nor the sceptical “we can never know”: it is a sharp intermediate statement. Vacuum energy is falsifiable in the directions the data cover, and a specific, mapped family of alternatives is forever indistinguishable from it.

2 The observability functional

Let $\mathcal{R}(k, a)$ denote the linear response of a dark-energy model—the perturbation it induces, as a function of comoving wavenumber k and scale factor a , in the quantities cosmology can measure. A cosmological constant has $\mathcal{R} \equiv 0$ in the perturbed sector and a fixed background; a dynamical model deviates from this in a pattern $\mathcal{R}(k, a)$ characteristic of its microphysics.

A probe P does not measure $\mathcal{R}$ pointwise. It measures a weighted projection: its sensitivity is concentrated on a subspace of the (k, a) response space, described by a kernel $W_P(k, a)$. We work on the response space L^2 of square-integrable functions on the $(\ln k, \ln a)$ domain with inner product

$$\langle \mathcal{R}_1, \mathcal{R}_2 \rangle = \int \mathcal{R}_1(k, a) \mathcal{R}_2(k, a) \mu(k, a) d \ln k d \ln a, \quad (1)$$

and induced norm $\|\mathcal{R}\|^2 = \langle \mathcal{R}, \mathcal{R} \rangle$. The weight $\mu(k, a)$ is

where survey geometry enters: in a full treatment it is the Fisher metric, $\mu \propto$ the inverse data covariance folded with the number of independent modes per cell (the cosmic-volume factor $\sim k^3$ within the survey footprint). In the schematic evaluation below we take μ flat on the logarithmic grid, which is the choice that makes the observable subspace depend only on the *support* of each probe kernel rather than on assumed noise levels; we return to this choice as a limitation. We define the observability of a feature $\mathcal{R}$ to the probe as

$$f_{\text{obs}}^{(P)}[\mathcal{R}] = \frac{\|\Pi_P \mathcal{R}\|}{\|\mathcal{R}\|}, \quad (2)$$

where Π_P is the orthogonal projection (in the inner product Eq. (1)) onto the span of the probe kernel. For a suite of probes $\{P_i\}$ the relevant projector is onto the combined span, $\Pi_{\text{probe}} = \Pi_{\text{span}\{W_{P_i}\}}$, and $f_{\text{obs}} \in [0, 1]$ measures the fraction of the feature’s response that lies within reach of the suite.

This is deliberately the information-geometric criterion made concrete: f_{obs} is the cosine between a feature and the observable subspace, and a Fisher analysis with the same kernels would assign a feature with $f_{\text{obs}} \rightarrow 0$ an unbounded confidence interval. The content of what follows is not the criterion—which is standard—but its *consequences* when the response space is that of dark energy and the projections are the specific ones the literature uses.

Three questions, one functional. The programme this paper closes answered three questions with what appear to be three different methods. Each is Eq. (2) under a particular Π_{probe} :

- A localised feature in the background expansion (a step or bump in the dark-energy density) is detectable iff its response is not projected out by the degeneracy directions of the distance–redshift data— Π is the nuisance-orthogonal Fisher projection onto the space orthogonal to $\{\Omega_m, \text{distance calibration}\}$.
- Horizon-entropy models are discriminable at the background level iff their responses are separated after the same projection—the discrimination question is Eq. (2) applied to differences of models.
- The structure-growth channel is closed for event-horizon models because their perturbation response is annihilated by the sub-horizon projection: the growth probe’s kernel is supported where a null-horizon response scales as $1/k^3$, so $\Pi_{\text{growth}}\mathcal{R} \rightarrow 0$.

The projections genuinely differ; the object projected, and the meaning of the norm, do not. This is the sense in which the three results are one.

3 The observable subspace is finite

Proposition 1 (Observable-subspace bound). *A suite of N probes with kernels $\{W_{P_i}\}$ renders observable only the N -dimensional (at most) subspace $\mathcal{O} = \text{span}\{W_{P_i}\}$ of the response space. Any feature $\mathcal{R} \in \mathcal{O}^\perp$ has $f_{\text{obs}}[\mathcal{R}] = 0$: it is unobservable to the suite for any exposure.*

The proposition is immediate—the span of N vectors has dimension at most N —but its force comes from the response space being infinite-dimensional. No finite programme closes it. Increasing exposure shrinks error bars *within* $\mathcal{O}$; it does not enlarge $\mathcal{O}$. Only a genuinely new probe, with a kernel outside the current span, adds a dimension.

The practical version replaces exact orthogonality with smallness. A feature with $f_{\text{obs}}[\mathcal{R}]$ near the value a random response direction would attain, $\sqrt{N/D}$ for a D -dimensional discretisation, is observationally indistinguishable from unobservable. It is this thresholded, quantitative form we now apply.

4 Can vacuum energy be confirmed?

We discretise the response space on a (k, a) grid spanning $k \in [10^{-4}, 1] h \text{ Mpc}^{-1}$ and $a \in [0.02, 1]$ ($D = 2400$ cells) and build physically-motivated kernels for six probes: background distance (supernovae and baryon acoustic oscillations), the growth rate $f\sigma_8$, the integrated Sachs–Wolfe effect, weak lensing, primary cosmic microwave background, and CMB lensing. Each kernel encodes where in (k, a) that probe has support—low redshift and sub-horizon for growth and lensing, very large scales for the ISW effect, high redshift for the CMB.

Two cautions must be stated at the outset, because they bound what the numbers below can mean. First, these kernels are *schematic*: analytic weight functions capturing the (k, a) support of each probe, not Fisher forecasts from survey specifications and a Boltzmann code. Second, treating each probe as a single basis vector *understates* the data: a real tomographic survey measures many redshift-binned spectra and spans a multi-dimensional subspace, not one direction. Both push the same way—a realistic suite has a larger observable subspace $\mathcal{O}$ than our six-vector caricature—so the specific overlap fractions we report are not to be read as forecasts. We therefore do not claim the decimal values; we claim the *ordering* they exhibit, and we test that the ordering is what survives.

It does. Expanding each probe into n_{bin} tomographic slices raises the subspace dimension (from 6 to ~ 36 at $n_{\text{bin}} = 10$) and lifts every overlap, but it lifts the low-redshift background direction to near unity while the blind directions rise far less and stay well below it: The

Table 1: Observability f_{obs} of a falsifiable direction (low- z background) and two blind directions as the observable subspace is enlarged by tomographic binning. The absolute values rise with dimension r , but the ordering (low- z overlap $\gg$ blind overlap) is preserved throughout. The random floor $\sqrt{r/D}$ is shown for reference.

n_{bin}	r	low- z bg	high- z DE	mid- z desert	floor
1	6	0.85	0.21	0.22	0.05
3	14	0.93	0.31	0.35	0.08
5	20	0.95	0.49	0.34	0.09
10	36	0.96	0.52	0.37	0.12

gap narrows as the data improve—as it must, since a richer suite sees more—but it does not close: at every dimensionality the low-redshift background direction is the most observable and the high-redshift and mid- z directions remain the least, closest to the floor. The qualitative result, a falsifiable set of directions cleanly separated from a blind set, is a property of *where the probes have support*, not of the caricature’s dimension. What a full Fisher forecast would change is how far into the blind zones a given survey generation reaches; what it would not change is that those zones are the last to be reached.

For definiteness we report the six-vector values below, with the explicit understanding that they are illustrative markers of the ordering, not forecasts; a random feature attains $f_{\text{obs}} \approx \sqrt{6/2400} \approx 0.05$ (the “blind” reference), each probe’s own kernel returns $f_{\text{obs}} = 1$, and a smooth physical response—being concentrated, not white—attains a higher baseline than a random one, which is why even the blind directions exceed 0.05 while remaining far below the observable ones.

A cosmological constant is not unfalsifiable: two ways of departing from it are well observed (Table 2). The low-redshift background direction is what current surveys target when they report a preference for evolving w ; the clustering direction reflects that a cosmological constant has *zero* perturbation response, so any dark-energy clustering is a large signal sitting squarely where growth and lensing probes look. If dark energy clusters at all, or if its background departs from $w = -1$ at low redshift, we can see that it is not Λ .

The opposite holds for an entire structured family of near-constant deviations (Table 3).

These sit at or near the blind floor. A dynamical dark energy that hugs $w = -1$ at low redshift and departs from vacuum energy only in these zones—early in time, on the largest scales, or in the redshift gap between the CMB and the low-redshift probes—produces a response almost entirely in $\mathcal{O}^\perp$. It is observationally indistinguishable from a cosmological constant, not because the effect is small but because the probes have no support where it lives.

Table 2: Directions in which the cosmological-constant hypothesis is falsifiable: deviations from $w = -1$ that a current-generation probe suite observes well. Values are the six-vector overlaps, illustrative of the ordering (see text); they rise toward unity under tomographic refinement (Table 1).

Deviation from Λ	f_{obs}
Low- z background evolution $w(a) \neq -1$	0.92
Presence of clustering ($c_s^2 < 1$, any)	0.87

Table 3: Directions in which the cosmological-constant hypothesis is not confirmable: near- Λ deviations that fall in the probe suite’s blind subspace. Values sit near the random floor and, while they rise under tomographic refinement, remain the least observable directions at every dimension (Table 1).

Deviation from Λ	f_{obs}
High- z / early dark energy ($z \gtrsim 4$)	0.21
Super-horizon ($k \lesssim 10^{-3}$)	0.45
Mid- z desert ($k \sim 10^{-3}$, $z \sim 2$ –4)	0.22

The bound. The two tables are the result. The cosmological-constant hypothesis is confirmable only in the observable subspace $\mathcal{O}$, and against the blind family it is not confirmable at all. The strongest claim the data can support is

dark energy is a cosmological constant within the observable subspace,

with the qualification carrying real content: it excludes a specific, enumerable family of dynamical models that will register in no experiment built from these six probes. “Dark energy is vacuum energy,” stated without that qualification, is not a conclusion cosmology can reach—not for want of precision, but for want of directions.

5 Discussion

The bound is epistemic, not dynamical: it says what is knowable about dark energy, not what dark energy is. It neither supports nor undermines the vacuum interpretation. It relocates the question. Confirming that dark energy is vacuum energy is not a matter of measuring w to more decimal places; it requires a probe with support in the blind zones—an observable sensitive to high-redshift or super-horizon dark-energy behaviour—without which a family of alternatives remains permanently degenerate with Λ .

Three limitations bound the claim. The probe kernels are schematic and each probe is treated as few basis vectors rather than a tomographic tower; a quantitative ver-

sion requires survey Fisher kernels with redshift binning, which will raise every overlap and enlarge the observable subspace. As Table 1 shows, this shifts the fractions substantially but preserves their ordering—the blind directions remain the least observable—so we claim the ordering and the existence of the blind family, not the decimal values. The response model for clustering is a Jeans form, not a Boltzmann output. And the framework addresses observability, not theoretical prior: Eq. (2) is the standard information criterion, and the contribution is its concrete consequence for the Λ question, not the criterion itself.

A word on scope is needed to avoid overstating the bound. The blind subspace is defined by *where in (k, a) the probes have support*, and that is a property of the data, not of any dark-energy model class. A deviation from $w = -1$ localised at high redshift, on super-horizon scales, or in the mid- z desert is hard to observe whatever its microphysical origin—geometric or scalar-field. So the blind family is not restricted to geometric dark energy; any dynamical model, including quintessence or k-essence, whose departure from Λ lives in those zones is equally degenerate with a cosmological constant. What *is* restricted to geometric dark energy is the clean *response-space bookkeeping* of the earlier sections, where the density perturbation is inherited from the metric and the projections have the simple form used in Sec. 2; scalar fields carry an independent kinetic term and a genuine Lagrangian sound speed, so their response space is larger and their placement in it requires the full perturbation treatment rather than the geometric shortcut. The epistemic bound—confirmation only within the observable subspace—holds for both; only the tidy geometric derivation is specific to the geometric class.

What survives all three is the shape of the answer. A finite cosmology sees a finite slice of what dark energy could be. The cosmological constant lives on the observable side of that slice, which is why it fits; a mapped family of its imitators lives on the blind side, which is why fitting is not confirming.

References

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